

Kundai Farai Sachikonye
Auf der Lichtung 19, 82178 Puchheim, Germany
github.com/fullscreen-triangle
kundai.sachikonye@bitspark.com
(+49) 1626386344

July 7, 2026

Clinical bioinformatician: multi-omics integration, disease-state modelling, and predictive stratification — reproducible, validated, and at population scale.

1 Profile

Computational biologist working on integrated multi-omics analysis and formal models of disease state across biological scales. I build reproducible, HPC-scale pipelines validated against authoritative reference data (gnomAD, UK Biobank, IEDB, NIST), and low-dimensional disease-vector models for multimorbidity, trajectory, and patient stratification. Fully authorised to work in Germany without restriction (permanent residence).

2 Education

2019–2021 PhD candidate, Computational Lipidomics Technische Universität München
Doctoral research in mass spectrometry-based lipidomics, multi-omics statistical modelling, and scientific computing. Departed programme to pursue independent computational research.

2018–2019 MSc Computational Biology Universität Tübingen

2014–2017 MSc Pharmaceutical Biotechnology HAW Hamburg

2011–2014 BSc Biochemistry and Cell Biology Jacobs University Bremen

3 Research Output

2025 Syndrome: Categorical Resolution of Disease Trajectories.

Disease state encoded as a vector $\mathbf{D} \in [0, 1]^8$ over eight cellular oscillator classes (protein, enzyme, channel, membrane, ATP, genetic expression, calcium, circadian); multimorbidity, trajectory, and stratification become geometry on a shared manifold. Intervention target $\arg \max_i w_i D_i$ as a sparse ℓ_1 LP with a priori side-effect bound; $\mathcal{O}(N \log N)$ trajectory resolution. *Manuscript*.

2025 On Disease Epistemology in Blind Receiver: Self-Consistency as the Unique Basis of Disease Detection.

Proves template-based diagnosis is epistemologically insufficient; derives loop-holonomy self-consistency as the unique viable criterion ($H_\ell \neq \text{Id}$ signals disease). Detects distributed, circuit-level inconsistency invisible to single-marker panels — relevant to gradual processes such as immune aging. AUC = 1.00 vs 0.896 for template comparison; 25/25 validation experiments pass. *Manuscript*.

2025 On the Thermodynamic Consequences of Categorical Completion on Biological Systems (adaptive immunity).

MHC binding affinity correlates with parent-protein categorical richness independently of classical sequence features ($r = 0.73$, $p < 10^{-12}$, $n = 2,847$ IEDB peptides).

Adding the term to NetMHCpan improves AUC $0.68 \rightarrow 0.81$ ($p < 0.001$, DeLong).
Manuscript.

4 Research Software

gospel Multi-omics integration pipeline: whole-genome VCF integrated with RNA-seq expression at 10^6 variants/second; Bayesian network inference, Gaussian mixture models, variational Bayes. Validated against 1000 Genomes Phase 3, gnomAD v3.1.2, UK Biobank. github.com/fullscreen-triangle/gospel

lavoisier Mass-spectrometry metabolomics framework; 98.9% feature-extraction accuracy against NIST reference libraries; PyTorch models, distributed computing (Ray), >100 GB mzML datasets via memory-mapped I/O. github.com/fullscreen-triangle/lavoisier

syndrome Implementation of the disease-vector framework: oscillator coherence computation, immunopeptidomics (MHC prediction, AUC 0.81 on IEDB), trajectory resolution. Python/Rust; browser-deployable tools. syndrome-seven.vercel.app

helicopter Multi-modal microscopy image analysis (fluorescence, brightfield, spectral) via sequential exclusion. github.com/fullscreen-triangle/helicopter

5 Research Experience

2019–2021 Computational Lipidomics — Bioinformatics & HPC Technische Universität München Munich High-throughput mass-spectrometry bioinformatics at HPC scale (>100 GB datasets): peak detection, ion database annotation, feature engineering, statistical modelling, multi-omics visualisation. Federated learning architecture for distributed analysis across clinical sites. Taught at Master's level.

2017–2018 Glycomics and Proteomics Universitätsklinikum Hamburg-Eppendorf Hamburg Automated container-based pipeline for standardised LC-MS/MS and MALDI-TOF characterisation; high-throughput processing and statistical QC.

2013 Cancer Biomarker Discovery University Medical Center Groningen Groningen UPLC-MS/MS shotgun proteomics protocol optimisation; high-dimensional data analysis for cervical cancer biomarker discovery.

2013 Pharmacokinetics Leibniz Lungenzentrum Borstel Quantification of matrix suppression in anti-TB drug efficacy measurement (Selected Reaction Monitoring LC-MS).

6 Technical Skills

Languages Python, R, Rust, TypeScript/JavaScript, Java, C, SQL, Bash

Multi-omics Variant calling (VCF), RNA-seq / transcriptome analysis, single-cell RNA-seq, mass-spectrometry metabolomics/proteomics (mzML, mzXML, BAM, FASTA), cross-omics integration, pathway analysis (KEGG, Reactome)

Statistics / ML Bayesian networks, Gaussian mixture models, variational Bayes, Gaussian processes, gradient boosting, graph neural networks, Transformers; PyTorch, scikit-learn, NumPy, SciPy, Pandas

Infrastructure Docker, distributed computing (Ray, Dask), HPC / SLURM, cloud (AWS, Azure), Git, PostgreSQL, Neo4j, Jupyter

Reference data 1000 Genomes, gnomAD, UK Biobank, IEDB, NIST

7 Languages

EnglishNative / Fluent (C2) GermanConversationalresident in Germany since 2011